If Zohran Mamdani can afford his policies with the Monte Carlo **simulation**

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Run thousands of simulations to list all the possible outcomes that Mamdani's proposal may cost

Parameters

```
grocery_mean = 60_000_000
grocery_std = grocery_mean * 0.2
```

```
free_bus_mean = 711_507_000
free_bus_std = 150_000_000
```

```
rent_freeze_mean = 6_840_000_000/4
rent_freeze_std = 400_000_000
```

```
landlord_prob = 0.05
landlord_mean = 1_500_000_000
landlord_std = 500_000_000
```

```
budget_threshold = 2_000_000_000
```

```
#How often his policies go above that amount or below that amount
```

Generate simulated costs for groceries, free bus, rent freeze, and potential landlord costs for a large number of iterations (e.g., 10,000 or 100,000) using their respective means and standard deviations, considering the landlord probability for conditional costs.

```
n_simulations = 100_000

grocery_costs = np.random.normal(grocery_mean, grocery_std, n_simulations)
free_bus_costs = np.random.normal(free_bus_mean, free_bus_std, n_simulations)
rent_freeze_costs = np.random.normal(rent_freeze_mean, rent_freeze_std, n_simulations)

landlord_costs = np.zeros(n_simulations)
landlord_random_numbers = np.random.rand(n_simulations)
landlord_impact_indices = landlord_random_numbers < landlord_prob
landlord_costs[landlord_impact_indices] = np.random.normal(landlord_mean, landlord_std, landlord_impact_
```

Simulated costs generated for groceries, free bus, rent freeze, and landlord.

```
total_costs = grocery_costs + free_bus_costs + rent_freeze_costs + landlord_costs
```

Total costs calculated for 100000 simulations.

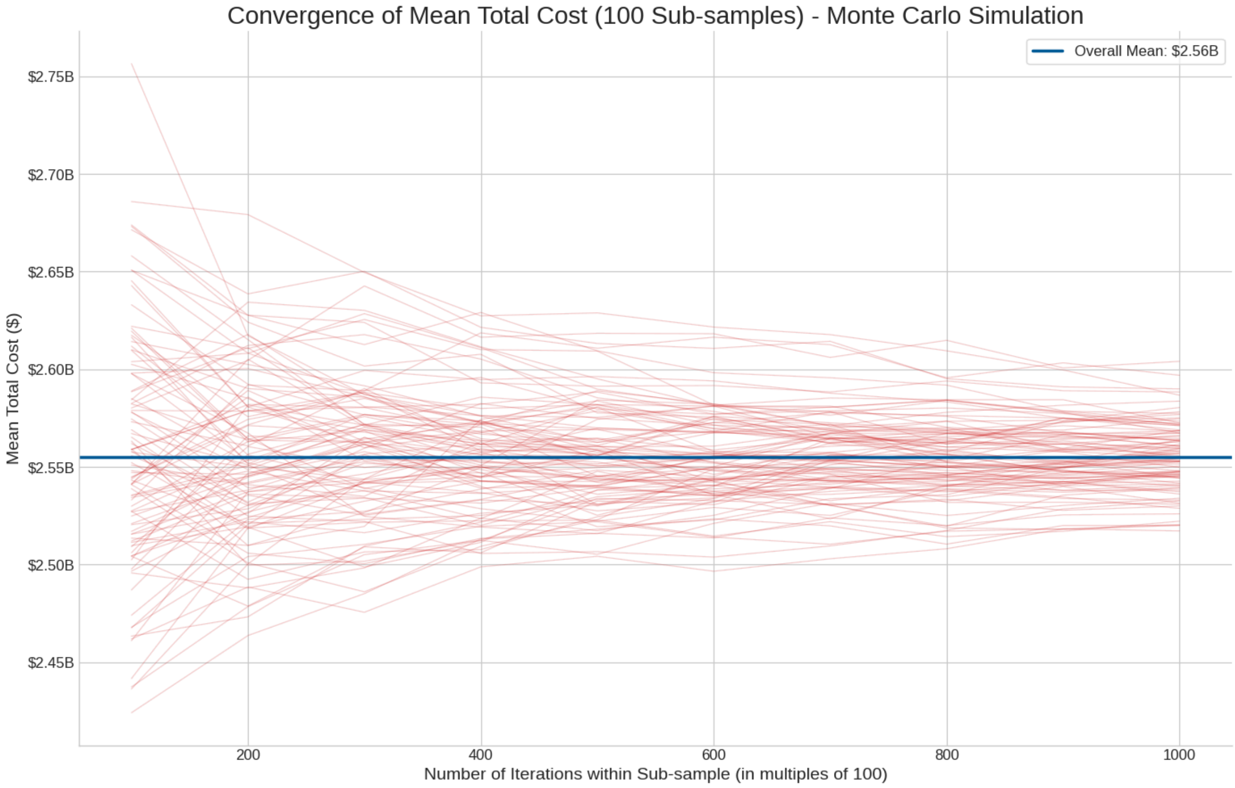
```
mean_total_cost = np.mean(total_costs)
std_total_cost = np.std(total_costs)

probability_exceeds_threshold = np.sum(total_costs > budget_threshold) / n_simulations
```

```
print(f"Mean total cost: ${mean_total_cost:,.2f}")
print(f"Standard deviation of total cost: ${std_total_cost:,.2f}")
print(f"Probability of total cost exceeding budget threshold (${budget_threshold:,.2f}): {probability_ex
```

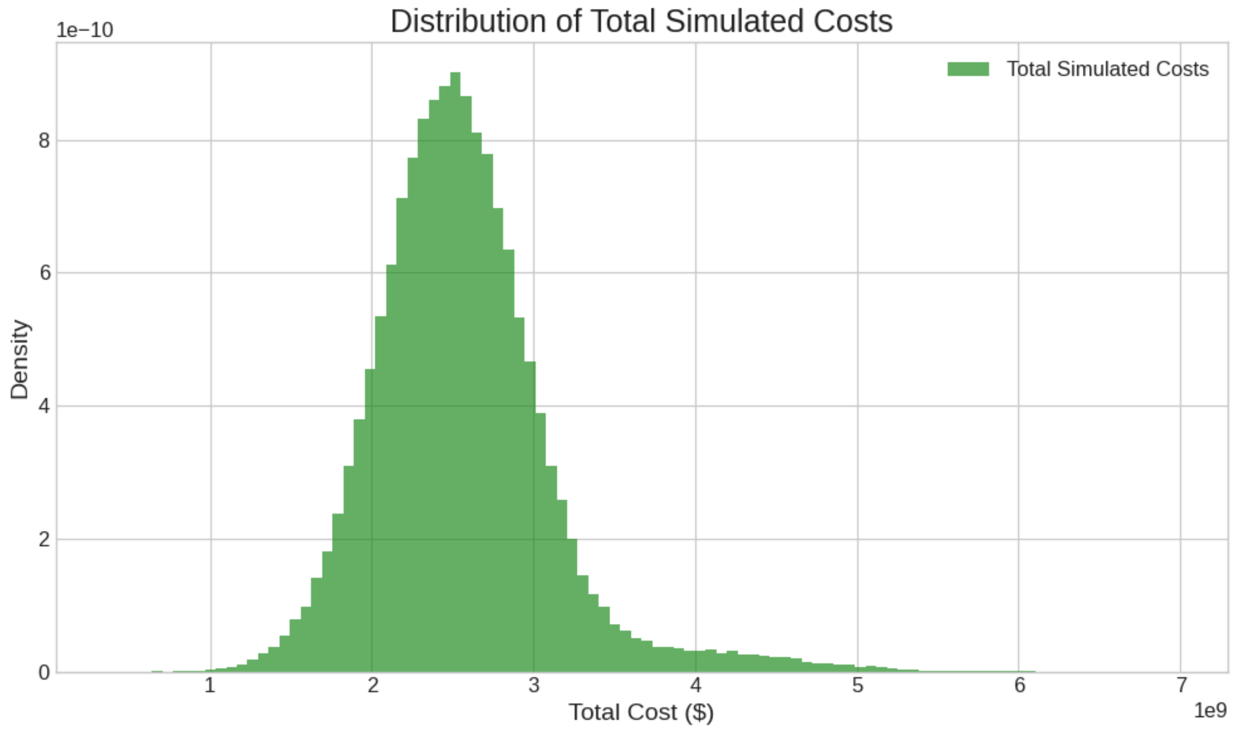
```
Mean total cost: $2,555,194,676.44
Standard deviation of total cost: $546,636,909.88
Probability of total cost exceeding budget threshold ($2,000,000,000.00): 87.62%
```

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import matplotlib.ticker as mticker
plt.style.use('seaborn-v0_8-whitegrid')

plt.rcParams['font.family'] = 'sans-serif'
# Removed: plt.rcParams['font.sans-serif'] = ['Arial', 'Helvetica', 'Lato'] # Prioritize common sans-ser
plt.rcParams['axes.titlesize'] = 18
plt.rcParams['axes.labelsize'] = 14
plt.rcParams['xtick.labelsize'] = 12
plt.rcParams['ytick.labelsize'] = 12
plt.rcParams['legend.fontsize'] = 12

def format_large_currency(amount, _pos=None):
    if abs(amount) >= 1_000_000_000:
        return f"${amount / 1_000_000_000:.2f}B"
    elif abs(amount) >= 1_000_000:
        return f"${amount / 1_000_000:.2f}M"
    elif abs(amount) >= 1_000:
        return f"${amount / 1_000:.2f}K"
    else:
        return f"${amount:,.1f}"

num_lines = 100
sims_per_line = n_simulations // num_lines
iterations_per_mean_point = 100
num_points_per_line = sims_per_line // iterations_per_mean_point

plt.figure(figsize=(14, 9))
x_axis = np.arange(iterations_per_mean_point, sims_per_line + 1, iterations_per_mean_point)

# Define professional color palette
line_color = '#CC0000'#005B9A' # A professional, darker blue
mean_color = '#005B9A' # A strong, clear red

for i in range(num_lines):
    segment_start_idx = i * sims_per_line
    segment_end_idx = (i + 1) * sims_per_line
    current_segment_costs = total_costs[segment_start_idx:segment_end_idx]

    mean_values_for_line = []
    for j in range(1, num_points_per_line + 1):
        current_iterations = j * iterations_per_mean_point
        mean_value = np.mean(current_segment_costs[:current_iterations])
        mean_values_for_line.append(mean_value)

    plt.plot(x_axis, mean_values_for_line, alpha=0.2, linewidth=0.8, color=line_color)

plt.axhline(y=mean_total_cost, color=mean_color, linewidth=2.5, label=f'Overall Mean: {format_large_curr

plt.title('Convergence of Mean Total Cost (100 Sub-samples) - Monte Carlo Simulation', fontsize=plt.rcPa
plt.xlabel(f'Number of Iterations within Sub-sample (in multiples of {iterations_per_mean_point})', font
plt.ylabel('Mean Total Cost ($)', fontsize=plt.rcParams['axes.labelsize'])

ax = plt.gca()
```

```
ax.yaxis.set_major_formatter(mticker.FuncFormatter(format_large_currency))

plt.legend(loc='upper right', frameon=True, edgecolor='lightgray')

ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(10, 6))
plt.hist(total_costs, bins=100, density=True, alpha=0.6, color='g', label='Total Simulated Costs')
plt.title('Distribution of Total Simulated Costs')
plt.xlabel('Total Cost ($)')
plt.ylabel('Density')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



Summary:

Q&A

- **What is the average total cost of Zohran Mamdani's policies?** The average total cost of Zohran Mamdani's policies, based on 100,000 Monte Carlo simulations, is $2,556,100,906.59.
- **What is the variability of the total cost?** The variability of the total cost, represented by the standard deviation, is $549,607,124.18.
- **What is the probability of exceeding the budget?** The probability that the total cost will exceed the budget threshold of $2,000,000,000.00 is 87.69%.

Data Analysis Key Findings

- The Monte Carlo simulation, conducted over 100,000 iterations, revealed an average total cost of approximately $2.56 billion for Zohran Mamdani's policies.
- The total cost shows significant variability, with a standard deviation of about $549.6 million, indicating a wide range of potential outcomes.
- There is a very high probability (87.69%) that the total cost of the policies will exceed the specified budget threshold of $2 billion.
- The histogram of total simulated costs visually confirms that a substantial portion of the simulated costs fall above the $2 billion budget threshold, with the distribution skewed towards higher costs.

Insights or Next Steps

- The high probability of exceeding the $2 billion budget threshold suggests that the current policy design, including the probabilities and cost distributions, is likely unsustainable within the set budget.
- Further analysis should focus on identifying which specific policy components (groceries, free bus, rent freeze, or landlord costs) contribute most significantly to the overall cost and variability to explore potential cost-reduction strategies or adjustments to policy parameters.